Chapter 4. Sampling Distributions and Large-Sample Theory

The Wishart distribution, distributions of $\bar{\mathbf{X}}$ and $\mathbf{S}$, MLE, consistency, multivariate CLT, assessing normality

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1 The Wishart Distribution

Motivation. The sample covariance matrix is random, and inference about $\boldsymbol{\Sigma}$ needs its distribution. The Wishart distribution is the matrix analogue of χ^2: a sum of outer products of independent normal vectors.

Definition 4.1: Wishart distribution

- The Wishart distribution is a family of distributions for symmetric positive definite matrices.
- The distribution of the $p \times p$ random matrix $\mathbf{M} = \mathbf{X}^\top \mathbf{X} = \sum_{i=1}^{n} \mathbf{X}_i \mathbf{X}_i^\top$ is a Wishart distribution.
- With m degrees of freedom,

$$W_p(m \mid \boldsymbol{\Sigma}) = \text{Wishart with } m \text{ d.f.} = \text{distribution of } \sum_{j=1}^{m} \mathbf{Z}_j \mathbf{Z}_j^\top, \qquad \mathbf{Z}_j \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

We also write $W_p(m, \boldsymbol{\Sigma})$.

- It is the multivariate extension of χ^2: if $\mathbf{M} \sim W_1(n, \sigma^2)$ then $\mathbf{M}/\sigma^2 \sim \chi^2_n$. E.g. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0, \sigma^2)$ gives $\mathbf{M} = \sum X_i^2$, $\mathbf{M}/\sigma^2 \sim \chi^2_n$.
- $W_p(n, \mathbf{I})$ is the *standard Wishart* distribution.
- If $\boldsymbol{\mu} = \mathbf{0}$ is known and $\mathbf{S} = \frac{1}{n}\mathbf{X}^\top \mathbf{X} = \frac{1}{n}\sum \mathbf{X}_i \mathbf{X}_i^\top$, then $n\mathbf{S} \sim W_p(n, \boldsymbol{\Sigma})$.
- [Supplement] If $m \geq p$ and $\boldsymbol{\Sigma} \succ 0$, $\mathbf{M}$ is PD with probability 1.

Theorem 4.2: properties of the Wishart distribution (Proposition 1)

(i) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ and $\mathbf{B}_{(p\times m)}$, $\mathbf{B}^\top \mathbf{M} \mathbf{B} \sim W_m(n, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, $\boldsymbol{\Sigma}^{-1/2} \mathbf{M} \boldsymbol{\Sigma}^{-1/2} \sim W_p(n, \mathbf{I}_p)$.

(iii) If $\mathbf{M}_i \sim W_p(n_i, \boldsymbol{\Sigma})$ $(i = 1, \ldots, k)$ are independent, $\sum_i \mathbf{M}_i \sim W_p(n, \boldsymbol{\Sigma})$ with $n = n_1 + \cdots + n_k$.

(iv) For $\mathbf{M}_n \sim W_p(n, \boldsymbol{\Sigma})$, $\mathrm{E}(\mathbf{M}_n) = n\boldsymbol{\Sigma}$ and $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$. Hence $\hat{\boldsymbol{\Sigma}}_n = \mathbf{M}_n/n$ has $\mathrm{E}(\hat{\boldsymbol{\Sigma}}_n) = \boldsymbol{\Sigma}$ and $\mathrm{Var}[(\hat{\boldsymbol{\Sigma}}_n)_{ij}] = (\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/n$.

(v) The law of large numbers and the Cramér–Wold device give $\mathbf{M}_n/n \to \boldsymbol{\Sigma}$ in probability as $n \to \infty$.

Proof. (i) Write $\mathbf{M} = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Then $\mathbf{B}^\top \mathbf{M} \mathbf{B} = \sum_i \mathbf{B}^\top \mathbf{x}_i (\mathbf{B}^\top \mathbf{x}_i)^\top$ and $\mathbf{B}^\top \mathbf{x}_i \overset{\text{iid}}{\sim} N_m(\mathbf{0}, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) Special case of (i) with $\mathbf{B} = \boldsymbol{\Sigma}^{-1/2}$.

(iii) With $\mathbf{M}_i = \sum_j \mathbf{x}_{ij} \mathbf{x}_{ij}^\top$, $\sum_i \mathbf{M}_i = \sum_i \sum_j \mathbf{x}_{ij} \mathbf{x}_{ij}^\top \sim W_p(\sum n_i, \boldsymbol{\Sigma})$.

(iv) $\mathrm{E}[\mathbf{M}_n] = \mathrm{E}(\sum_k \mathbf{x}_k \mathbf{x}_k^\top) = \sum_k \mathrm{Var}(\mathbf{x}_k) = n\boldsymbol{\Sigma}$. Terms with different k are independent, so $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n\{\mathrm{E}(X_{ki}^2 X_{kj}^2) - \Sigma_{ij}^2\}$. Using $\mathrm{E}(X_{ki} \mid X_{kj}) = \frac{\Sigma_{ij}}{\Sigma_{jj}} X_{kj}$ and $X_{ki} - \mathrm{E}(X_{ki} \mid X_{kj}) \perp\!\!\!\perp X_{kj}$,

$$\mathrm{E}[(X_{ki}X_{kj})^2] = \mathrm{E}[X_{kj}^2 \mathrm{E}(X_{ki}^2 \mid X_{kj})] = \frac{\Sigma_{ij}^2}{\Sigma_{jj}^2} 3\Sigma_{jj}^2 + \left(\Sigma_{ii} - \frac{\Sigma_{ij}^2}{\Sigma_{jj}}\right)\Sigma_{jj} = 2\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj},$$

so $\mathrm{Var} = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$.

(v) By Cramér–Wold it suffices that $\sum_{ij} a_{ij} M_{ij}/n \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$ for all a_{ij}. Since $\mathbf{M}_n/n = \frac{1}{n}\sum_k \mathbf{x}_k\mathbf{x}_k^\top$ is an average of iid random matrices, the LLN gives $\sum_{ij} a_{ij}\frac{M_{ij}}{n} \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$. ■

2 Wishart Quadratic Forms and the Distribution of S

Motivation. Projections of a Wishart matrix are again Wishart or χ^2, which yields the distribution of $\mathbf{S}$ and its independence from $\bar{\mathbf{X}}$, just as in the univariate case.

Theorem 4.3: Wishart quadratic forms (Lemma 5)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_p(\mathbf{0}, \mathbf{\Sigma})$, $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^\top$, and $\mathbf{A}$ an $n \times n$ real symmetric matrix. Then

$$\mathbf{X}^\top \mathbf{A} \mathbf{X} \sim W_p(r, \mathbf{\Sigma}) \iff \mathbf{A} \text{ is idempotent with } \mathrm{rank}(\mathbf{A}) = r.$$

Partial proof ($\Leftarrow$). Since $\mathbf{A}$ is symmetric idempotent of rank r, $\mathbf{A} = \mathbf{\Gamma}\mathbf{\Lambda}\mathbf{\Gamma}^\top$ with $\mathbf{\Lambda} = \mathrm{diag}(1, \dots, 1, 0, \dots, 0)$. Write $\mathbf{A} = \mathbf{P}\begin{pmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}\mathbf{P}^\top$ for an $n \times n$ orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$, so $\mathbf{A} = \sum_{j=1}^r \mathbf{p}_j\mathbf{p}_j^\top$ and

$$\mathbf{X}^\top \mathbf{A} \mathbf{X} = \sum_{j=1}^{r} \mathbf{X}^\top \mathbf{p}_j \mathbf{p}_j^\top \mathbf{X}.$$

Now $\mathbf{X}^\top \mathbf{p}_j = \sum_{i=1}^n p_{ij}\mathbf{X}_i \sim N_p(\mathbf{0}, \mathbf{\Sigma})$, since

$$\mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i\Big) = \mathbf{\Sigma} \cdot \mathbf{p}_j^\top \mathbf{p}_j = \mathbf{\Sigma}, \qquad \mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i, \sum_i p_{ik}\mathbf{X}_i\Big) = \mathbf{p}_j^\top \mathbf{p}_k \mathbf{\Sigma} = \mathbf{0} \ (j \neq k);$$

being jointly normal, $\mathbf{z}_j = \mathbf{X}^\top \mathbf{p}_j$ are iid $N_p(\mathbf{0}, \mathbf{\Sigma})$. Hence $\mathbf{X}^\top \mathbf{A} \mathbf{X} = \sum_{j=1}^r \mathbf{z}_j\mathbf{z}_j^\top \sim W_p(r, \mathbf{\Sigma})$. ([Supplement] $\Rightarrow$ reduces to Theorem 3.21 with $p = 1$: $\mathbf{a}^\top \mathbf{X}^\top \mathbf{A}\mathbf{X}\mathbf{a} \sim (\mathbf{a}^\top \mathbf{\Sigma}\mathbf{a})\chi_r^2$ for every $\mathbf{a}$.) ■

Theorem 4.4: distribution of the sample covariance matrix; $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$

If $\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, then

$$\bar{\mathbf{X}}_n \sim N_p(\boldsymbol{\mu}, \tfrac{1}{n}\mathbf{\Sigma}), \qquad (n-1)\mathbf{S} = \mathbf{X}^\top\Big(\mathbf{I}_n - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\Big)\mathbf{X} \sim W_p(n-1, \mathbf{\Sigma}), \qquad \bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}.$$

Proof.

Step 1 ($\bar{\mathbf{X}}$). $\bar{\mathbf{X}} = \frac{1}{n}\mathbf{X}^\top \mathbf{1}_n$ is a linear transformation of the jointly normal $(\mathbf{X}_1^\top, \dots, \mathbf{X}_n^\top)^\top$ with mean $\boldsymbol{\mu}$ and covariance $\frac{1}{n^2}\sum_i \mathbf{\Sigma} = \frac{1}{n}\mathbf{\Sigma}$, so $\bar{\mathbf{X}} \sim N_p(\boldsymbol{\mu}, \frac{1}{n}\mathbf{\Sigma})$ by Theorem 3.8.

Step 2 ($\mathbf{S}$). $\mathbf{S} = \frac{1}{n-1}\mathbf{X}_C^\top \mathbf{X}_C = \frac{1}{n-1}\mathbf{X}^\top(\mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)\mathbf{X}$ with $\mathbf{X}_C = \mathbf{H}_n\mathbf{X}$. The matrix $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$ is symmetric idempotent of rank $n-1$. Since it annihilates $\mathbf{1}_n$, $\mathbf{X}^\top \mathbf{H}_n \mathbf{X} = \mathbf{Y}^\top \mathbf{H}_n \mathbf{Y}$ with $\mathbf{Y} = \mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top$, whose rows are iid $N_p(\mathbf{0}, \mathbf{\Sigma})$, and Theorem 4.3 gives $(n-1)\mathbf{S} \sim W_p(n-1, \mathbf{\Sigma})$.

Step 3 (independence) [Supplement]. Choose an orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$ with $\mathbf{p}_1 = \mathbf{1}_n/\sqrt{n}$ and let $\mathbf{z}_j = (\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)^\top \mathbf{p}_j$. As above $\mathbf{z}_1, \dots, \mathbf{z}_n \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \mathbf{\Sigma})$, $\mathbf{z}_1 = \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})$, and since $\mathbf{H}_n = \sum_{j=2}^n \mathbf{p}_j\mathbf{p}_j^\top$, $(n-1)\mathbf{S} = \sum_{j=2}^n \mathbf{z}_j\mathbf{z}_j^\top$. The first depends only on $\mathbf{z}_1$, the second only on $\mathbf{z}_2, \dots, \mathbf{z}_n$. ■

3 A Theorem on Inverse Wishart Ratios

Motivation. Hotelling's T^2 involves $\mathbf{S}^{-1}$. This theorem reduces quadratic forms in an inverse Wishart matrix to ratios of χ^2 variables, which is the key step toward the F distribution of T^2.

Theorem 4.5: ratios involving an inverse Wishart matrix (Theorem 3)

Suppose $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$, $\mathbf{a} \in \mathbb{R}^p$, and $n > p-1$. Then

(i) $(\boldsymbol{\Sigma}^{-1})_{11}/(\mathbf{M}^{-1})_{11} \sim \chi^2_{n-p+1}$;

(ii) for any constant $\mathbf{a} \neq \mathbf{0}$, $\dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top \mathbf{M}^{-1}\mathbf{a}} \sim \chi^2_{n-p+1}$;

(iii) if $\mathbf{y} \in \mathbb{R}^p$ is random and independent of $\mathbf{M}$ (with $P(\mathbf{y}=\mathbf{0})=0$), then $\dfrac{\mathbf{y}^\top \boldsymbol{\Sigma}^{-1}\mathbf{y}}{\mathbf{y}^\top \mathbf{M}^{-1}\mathbf{y}} \sim \chi^2_{n-p+1}$ and is independent of $\mathbf{y}$.

Sketch. (i) is a special case of (ii), but we prove (i) first and generalize it to obtain (ii); (ii) implies (iii), since the conditional distribution of $\mathbf{y}^\top\boldsymbol{\Sigma}^{-1}\mathbf{y}/\mathbf{y}^\top\mathbf{M}^{-1}\mathbf{y}$ given $\mathbf{y}=\mathbf{a}$ is χ^2_{n-p+1} for almost all $\mathbf{a}$ (i.e. $\mathbf{a}\neq\mathbf{0}$) by (ii).

Proof. (i) Write

$$\mathbf{M} = \begin{pmatrix} m_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \boldsymbol{\sigma}_{12} \\ \boldsymbol{\sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

By the block-inverse formula,

$$\frac{(\boldsymbol{\Sigma}^{-1})_{11}}{(\mathbf{M}^{-1})_{11}} = \frac{m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21}}{\sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}}.$$

Write $\mathbf{M} = \mathbf{X}^\top\mathbf{X} = \sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ ($\mathbf{x}_1$: first column). Then

$$m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21} = \mathbf{x}_1^\top\mathbf{x}_1 - \mathbf{x}_1^\top\mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{x}_1 = \mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1,$$

$$\mathbf{A} = \mathbf{A}(\mathbf{X}_2) = \mathbf{I}_n - \mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top.$$

Given $\mathbf{X}_2$, $\mathbf{x}_1 \mid \mathbf{X}_2 \sim N_n(\mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21},\ \sigma_{11\cdot 2}\mathbf{I}_n)$, $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$. Since $\mathbf{A}\mathbf{X}_2 = \mathbf{0}$, $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 = (\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})^\top\mathbf{A}(\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})$. For any given $\mathbf{X}_2$, $\mathbf{A}$ is symmetric idempotent with $\operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A}) = n - \operatorname{tr}\{(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{X}_2\} = n-(p-1)$, so Theorem 3.21 gives

$$\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \mid \mathbf{X}_2 \sim \sigma_{11\cdot 2}\,\chi^2_{n-p+1}.$$

This does not depend on $\mathbf{X}_2$, so marginally $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \sim \sigma_{11\cdot 2}\chi^2_{n-p+1}$, independent of $\mathbf{X}_2$. Hence the ratio is $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$.

(i)⇒(ii) Given $\mathbf{a}$, take a $p\times p$ orthogonal $\mathbf{P}$ whose first column is $\mathbf{a}/\|\mathbf{a}\|$, i.e. $\mathbf{a}^\top\mathbf{P} = \mathbf{e}_1^\top$ (w.l.o.g. $\|\mathbf{a}\| = 1$). Then

$$\frac{\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}} = \frac{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}}{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}} = \frac{\mathbf{e}_1^\top\boldsymbol{\Sigma}_{\mathbf{P}}^{-1}\mathbf{e}_1}{\mathbf{e}_1^\top\mathbf{M}_{\mathbf{P}}^{-1}\mathbf{e}_1}, \quad \boldsymbol{\Sigma}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P},\ \mathbf{M}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}.$$

By Theorem 4.2, $\mathbf{M}_{\mathbf{P}} = \mathbf{P}^\top\mathbf{M}\mathbf{P} \sim W_p(n, \boldsymbol{\Sigma}_{\mathbf{P}})$, so (i) applies: $(\boldsymbol{\Sigma}_{\mathbf{P}}^{-1})_{11}/(\mathbf{M}_{\mathbf{P}}^{-1})_{11} \sim \chi^2_{n-p+1}$.

(ii)⇒(iii) Conditioning on $\mathbf{y}=\mathbf{a}$ leaves the law of $\mathbf{M}$ unchanged (independence), so by (ii) the conditional distribution is χ^2_{n-p+1} for almost all $\mathbf{a}\neq\mathbf{0}$. It does not depend on $\mathbf{a}$, so it is also the marginal law and is independent of $\mathbf{y}$. ∎

Remark 4.6: residuals and projections in multivariate analysis

1. The residual of $\mathbf{x}_1$ after projecting onto the span of $\mathbf{X}_2$ is $\mathbf{x}_1^* = \mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathbf{x}_1 - \mathrm{BLP}(\mathbf{x}_1 \mid \mathbf{X}_2)$.
2. $\mathbf{x}_1^*$ is orthogonal to (uncorrelated with) $\mathbf{X}_2$; being Gaussian, they are independent.
3. The partial variance $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathrm{Var}(\mathbf{x}_1^*)$ is the variance left after removing $\mathbf{X}_2$.

The residual sum of squares has $n-(p-1)$ degrees of freedom. This theorem is the key to the F distribution of Hotelling's T^2 (Ch. 5).

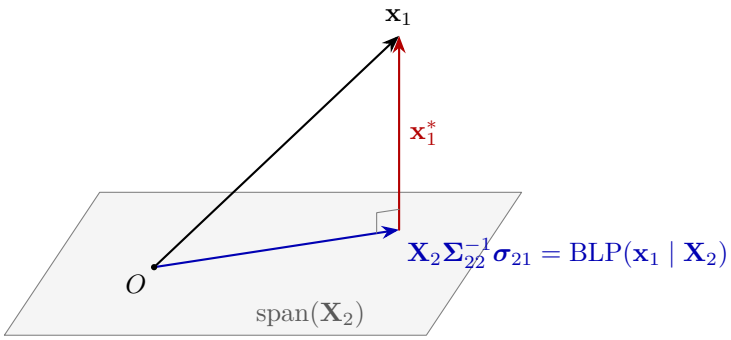


Figure. *Residual and projection. The residual $\mathbf{x}_1^*$ is what is left of $\mathbf{x}_1$ after projecting it onto the span of $\mathbf{X}_2$; it is orthogonal to that span, and $\sigma_{11\cdot 2} = \mathrm{Var}(\mathbf{x}_1^*)$ measures its size.*

4 Maximum Likelihood Estimation

Motivation. Under normality the maximum likelihood estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the sample mean and the (biased) sample covariance. The proof uses a trace inequality rather than differentiation.

Theorem 4.7: MLE of μ and Σ (Result 4.11)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be iid $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p$. The MLEs are

$$\hat{\boldsymbol{\mu}}_{\mathrm{MLE}} = \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i, \qquad \hat{\boldsymbol{\Sigma}}_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top =: \mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}.$$

As random quantities, $\bar{\mathbf{X}}$ and $\frac{1}{n}\sum_j (\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top$ are the maximum likelihood *estimators*; their observed values are the maximum likelihood *estimates*.

The MLE jointly minimizes the negative log-likelihood

$$\begin{aligned}\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \mathbf{x}^n) &:= -\log L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) + c \\ &= \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{n}{2}(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) + c.\end{aligned}$$

(Expand $\mathbf{x}_i - \boldsymbol{\mu} = (\mathbf{x}_i - \bar{\mathbf{x}}) + (\bar{\mathbf{x}} - \boldsymbol{\mu})$; the cross term vanishes.)

Step 1 (profile out μ). ℓ depends on $\boldsymbol{\mu}$ only through $(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \geq 0$, minimized by $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every $\boldsymbol{\Sigma}$. With $\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathrm{tr}\{\boldsymbol{\Sigma}^{-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\} = n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$,

$$2\ell(\hat{\boldsymbol{\mu}}_{\mathrm{MLE}}, \boldsymbol{\Sigma}) = n\log|\boldsymbol{\Sigma}| + n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) + c,$$

so it remains to minimize $f(\boldsymbol{\Sigma}) = \log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$ over $\boldsymbol{\Sigma} \succ 0$.

First proof: von Neumann's inequality. Decompose $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ and $\mathbf{S}_0 = \mathbf{V}\mathbf{D}\mathbf{V}^\top$ (λ_i, d_i in decreasing order; $d_i > 0$ a.s. when $n > p$). The eigenvalues of $\boldsymbol{\Sigma}^{-1}$ in increasing order are $1/\lambda_1 \le \cdots \le 1/\lambda_p$, so von Neumann's lower bound gives

$$\log|\boldsymbol{\Sigma}| + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) \ge \log(\lambda_1 \cdots \lambda_p) + \sum_{i=1}^{p} \frac{d_i}{\lambda_i} = \sum_{i=1}^{p} \Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big),$$

with equality iff $\mathbf{U} = \mathbf{V}$. Let $a_i = d_i/\lambda_i$. Then

$$\sum_{i=1}^{p} \Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big) = \sum_{i=1}^{p} (\log d_i + a_i - \log a_i) = \log|\mathbf{S}_0| + \sum_{i=1}^{p} (a_i - \log a_i) \ge \log|\mathbf{S}_0| + p,$$

since $x - \log x \ge 1$. The bound is attained iff (1) $\mathbf{U} = \mathbf{V}$ and (2) $a_i = 1$ for all i, i.e. $\boldsymbol{\Lambda} = \mathbf{D}$: $\hat{\boldsymbol{\Sigma}}_{\text{MLE}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top = \mathbf{V}\mathbf{D}\mathbf{V}^\top = \mathbf{S}_0$. ■

Second proof: matrix calculus. Reparametrise by the precision matrix $\mathbf{K} = \boldsymbol{\Sigma}^{-1}$ ($\boldsymbol{\Sigma} \mapsto \boldsymbol{\Sigma}^{-1}$ is a bijection of the PD cone, so by the change-of-variables part of Theorem 2.26 the optimizers correspond). $g(\mathbf{K}) = -\log|\mathbf{K}| + \operatorname{tr}(\mathbf{K}\mathbf{S}_0)$ is defined on the open convex set $\{\mathbf{K} \succ 0\}$, and for symmetric $\mathbf{H} \neq \mathbf{0}$

$$D^2 g(\mathbf{K})[\mathbf{H}] = \operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}\mathbf{K}^{-1}\mathbf{H}) = \|\mathbf{K}^{-1/2}\mathbf{H}\mathbf{K}^{-1/2}\|_F^2 > 0,$$

so g is strictly convex (Theorem 2.25). The first-order condition

$$Dg(\mathbf{K})[\mathbf{H}] = -\operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}) + \operatorname{tr}(\mathbf{S}_0\mathbf{H}) = \operatorname{tr}\{(\mathbf{S}_0 - \mathbf{K}^{-1})\mathbf{H}\} = 0 \quad \forall \mathbf{H}$$

and Theorem 2.24 give $\mathbf{K}^{-1} = \mathbf{S}_0$, the unique minimizer. By profiling (Theorem 2.26), $(\bar{\mathbf{x}}, \mathbf{S}_0)$ is the joint minimizer. ■

Remark 4.8: invariance and the maximized likelihood

[Supplement] A function of the MLE is the MLE of the function: e.g. the eigenvalues of $\mathbf{S}_0$ estimate those of $\boldsymbol{\Sigma}$, and r_{ij} is the MLE of ρ_{ij}. The maximum is $L(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}) = (2\pi)^{-np/2}|\mathbf{S}_0|^{-n/2}e^{-np/2}$, which depends on the data only through the generalized variance $|\mathbf{S}_0|$; this is used for likelihood ratio tests (Ch. 5).

5 Large-Sample Behavior of $\bar{\mathbf{X}}$ and $\mathbf{S}$

Motivation. Without normality, the CLT still makes $\bar{\mathbf{X}}$ approximately normal and $\mathbf{S}$ consistent, so normal-theory procedures remain valid for large samples.

Theorem 4.9: law of large numbers (Result 4.12)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_j) = \boldsymbol{\mu}$ and finite $\mathrm{Cov}(\mathbf{X}_j) = \boldsymbol{\Sigma}$. Then, as $n \to \infty$,

$$\text{(a) } \bar{\mathbf{X}}_n \xrightarrow{p} \boldsymbol{\mu}, \qquad \text{(b) } \mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}, \qquad \text{(c) } \mathbf{S}_0 \xrightarrow{p} \boldsymbol{\Sigma}.$$

Moreover $\bar{\mathbf{X}}_n$ and $\mathbf{S}$ are unbiased.

Proof. (a) By Markov's inequality,

$$P(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\| > \epsilon) \le \frac{1}{\epsilon^2}\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2 = \frac{1}{\epsilon^2}\mathrm{E}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) = \frac{1}{\epsilon^2}\mathrm{E}[\operatorname{tr}\{(\bar{\mathbf{X}}_n - \boldsymbol{\mu})(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\}]$$

$$= \frac{1}{\epsilon^2}\mathrm{tr}(\mathrm{Var}(\bar{\mathbf{X}}_n)) = \frac{\mathrm{tr}(\boldsymbol{\Sigma})}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0.$$

(b) Adding and subtracting $\boldsymbol{\mu}$ inside each factor and using $\sum_i(\mathbf{X}_i - \boldsymbol{\mu}) = n(\bar{\mathbf{X}} - \boldsymbol{\mu})$,

$$(n-1)\mathbf{S} = \sum_{i=1}^{n}(\mathbf{X}_i - \bar{\mathbf{X}}_n)(\mathbf{X}_i - \bar{\mathbf{X}}_n)^\top = \sum_{i=1}^{n}(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top.$$

Hence

$$\frac{n-1}{n}\mathbf{S} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - (\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \xrightarrow{p} \boldsymbol{\Sigma} - \mathbf{0} = \boldsymbol{\Sigma}.$$

The first term is an average of iid matrices with mean $\mathrm{E}(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top = \boldsymbol{\Sigma}$, so it converges to $\boldsymbol{\Sigma}$ by the weak LLN applied entrywise (matrix convergence is entrywise, Theorem 2.30). The second term tends to $\mathbf{0}$ by (a) and the CMT ($\mathbf{x} \mapsto \mathbf{x}\mathbf{x}^\top$ is continuous). Since $n/(n-1) \to 1$, $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$.

(c) $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, which was shown to converge to $\boldsymbol{\Sigma}$ in (b).

Unbiasedness. $\mathrm{E}\bar{\mathbf{X}}_n = \boldsymbol{\mu}$. Under normality $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$, so $\mathrm{E}\{(n-1)\mathbf{S}\} = (n-1)\boldsymbol{\Sigma}$ by Theorem 4.2(iv), i.e. $\mathrm{E}\mathbf{S} = \boldsymbol{\Sigma}$. [Supplement] Without normality, taking expectations in the identity of (b) gives the same, $\mathrm{E}\{(n-1)\mathbf{S}\} = n\boldsymbol{\Sigma} - n\mathrm{Var}(\bar{\mathbf{X}}_n) = (n-1)\boldsymbol{\Sigma}$; under normality Theorem 4.2(iv) also gives the rate $\mathrm{E}\|\mathbf{S} - \boldsymbol{\Sigma}\|_F^2 = \sum_{ij}(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/(n-1) \to 0$. ■

Remark 4.10: rate of the estimation error

With the Frobenius norm $\|\mathbf{A}\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$, the estimation errors are $E_{\bar{u}} = \|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2$, $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$ and $E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$ ($\mathbf{R}_0$ the population correlation matrix). [Supplement] $\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2 = \mathrm{tr}\boldsymbol{\Sigma}/n$, so $E_{\bar{u}} = \|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2$ decays at rate $n^{-1/2}$; so does $\|\mathbf{S} - \boldsymbol{\Sigma}\|_F$ under normality. On a log–log plot of $E_{\bar{u}}, E_S, E_R$ against n the slope is about $-1/2$.

Theorem 4.11: multivariate central limit theorem (Result 4.13(i))

Let $\mathbf{X}_1, \mathbf{X}_2, \cdots \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_i) = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{X}_i) = \boldsymbol{\Sigma}$. Then

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(\mathbf{X}_i - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \text{equivalently} \qquad \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Proof. By Cramér–Wold it suffices to show $\mathbf{t}^\top\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow \mathbf{t}^\top\mathbf{Y}$, $\mathbf{Y} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, for every $\mathbf{t}$. $W_i = \mathbf{t}^\top(\mathbf{X}_i - \boldsymbol{\mu})$ are iid scalars with mean 0 and variance $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t}$, so the univariate CLT gives $\sqrt{n}\bar{W}_n \Rightarrow N(0, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$, the law of $\mathbf{t}^\top\mathbf{Y}$ (if $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t} = 0$, $W_i = 0$ a.s. and the claim is trivial). ■

Theorem 4.12: large-sample chi-square (Result 4.13(ii))

Under the same assumptions with $\boldsymbol{\Sigma} \succ 0$, $P(\mathbf{S} \text{ invertible}) \to 1$ and $T_n := n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top r(\mathbf{S})(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow \chi_p^2$, where $r(\mathbf{A}) := \mathbf{A}^{-1}$ if $|\mathbf{A}| \neq 0$ and $r(\mathbf{A}) := \mathbf{I}_p$ otherwise; thus $T_n = n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ whenever $\mathbf{S}$ is invertible.

Proof. Let $\mathbf{Z}_n := \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ and $\mathbf{Z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$.

(i) $\mathbf{Z}_n \Rightarrow \mathbf{Z}$. For every $\mathbf{t} \in \mathbb{R}^p$, $\mathbf{t}^\top\mathbf{Z}_n = \sqrt{n}(\mathbf{t}^\top\bar{\mathbf{X}}_n - \mathbf{t}^\top\boldsymbol{\mu}) \Rightarrow N(0, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$ by the univariate CLT, so $\mathbf{Z}_n \Rightarrow \mathbf{Z}$ by Cramér–Wold (this is Theorem 4.11).

(ii) Handling a possibly singular S. Let $A_n := \{\mathbf{S} \text{ is invertible}\}$. Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$ (Theorem 4.9(b)) and $\mathbf{A} \mapsto |\mathbf{A}|$ is continuous, $|\mathbf{S}| \xrightarrow{p} |\boldsymbol{\Sigma}| > 0$, hence

$$P(A_n) \geq P(|\mathbf{S}| \geq \tfrac{1}{2}|\boldsymbol{\Sigma}|) \longrightarrow 1.$$

Define $r(\mathbf{A}) = \mathbf{A}^{-1}$ if $|\mathbf{A}| \neq 0$ and $r(\mathbf{A}) = \mathbf{I}_p$ otherwise. By the definition of r,

$$T_n = n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n} + n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top (\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n^c} = \mathbf{Z}_n^\top r(\mathbf{S}) \mathbf{Z}_n,$$

and T_n coincides with the statistic of the theorem on A_n, an event of probability tending to one.

(iii) $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$**.** Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$, by the CMT it suffices that r is continuous at the point $\boldsymbol{\Sigma}$. Since $|\boldsymbol{\Sigma}| > 0$ and $\{\mathbf{A} : |\mathbf{A}| \neq 0\}$ is open, $r(\mathbf{A}) = \mathbf{A}^{-1} = \mathrm{adj}(\mathbf{A})/|\mathbf{A}|$ on a neighborhood of $\boldsymbol{\Sigma}$ (Section 1 of Ch. 2), whose entries are polynomials divided by a nonvanishing polynomial; so r is continuous at $\boldsymbol{\Sigma}$ and $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$.

(iv) Conclusion. Let $Q(\mathbf{z}, \mathbf{A}) = \mathbf{z}^\top \mathbf{A} \mathbf{z}$, so $T_n = Q(\mathbf{Z}_n, r(\mathbf{S}))$. By Slutsky, $(\mathbf{Z}_n, r(\mathbf{S})) \Rightarrow (\mathbf{Z}, \boldsymbol{\Sigma}^{-1})$, and Q is continuous, so by the CMT

$$T_n = Q(\mathbf{Z}_n, r(\mathbf{S})) \Rightarrow Q(\mathbf{Z}, \boldsymbol{\Sigma}^{-1}) = \mathbf{Z}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Z} \sim \chi^2_p$$

(Example 3.23). ■

This is the basis of large-sample tests and confidence regions for the mean without normality (Theorems 5.16 and 5.17).

Theorem 4.13: asymptotic normality of $h(\bar{\mathbf{X}}_n)$ (delta method)

For $h : \mathbb{R}^p \to \mathbb{R}$ differentiable at $\boldsymbol{\mu}$ with gradient $\nabla h(\boldsymbol{\mu})$,

$$\sqrt{n}\{h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})\} \Rightarrow N_p(0,\ (\nabla h(\boldsymbol{\mu}))^\top \boldsymbol{\Sigma} (\nabla h(\boldsymbol{\mu}))).$$

Proof. Taylor: $h(\bar{\mathbf{X}}_n) = h(\boldsymbol{\mu}) + (\nabla h(\boldsymbol{\mu}))^\top (\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2)$. Multiply by $\sqrt{n}$: $\sqrt{n}(h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})) = (\nabla h(\boldsymbol{\mu}))^\top \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\sqrt{n}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2)$, and the remainder is $o_p(1)$. Slutsky gives $\Rightarrow (\nabla h(\boldsymbol{\mu}))^\top N_p(\mathbf{0}, \boldsymbol{\Sigma})$. ■

6 Assessing the Assumption of Normality

Motivation. Most procedures in later chapters assume $\mathbf{X}_j \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Normality itself cannot be verified, but its consequences can be checked one at a time: normal marginals, elliptical pairs and χ^2_p distances.

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

- every $\mathbf{a}^\top \mathbf{X}$ is normal (Result 4.2), so each marginal X_i is normal;
- the density contours are the ellipsoids $(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$, so each pair (X_i, X_k) has elliptical contours;
- the squared distance $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$ (Result 4.7): very large distances are rare.

These are necessary conditions only: a violation rules out normality, but passing them does not prove it. The checks are: (1) normal marginals: proportions, Q–Q plot, r_Q; (2) elliptical pairs: the 50% contour; (3) overall fit and outliers: the chi-square plot.

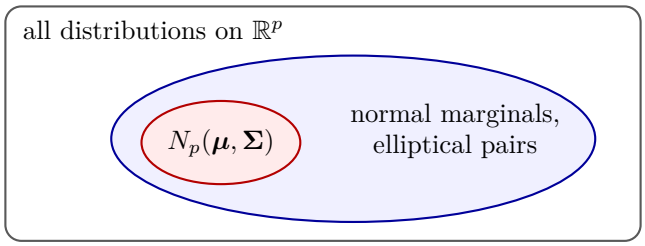


Figure. *The checks below test necessary conditions. Data from a non-normal law in the blue region pass the marginal and pairwise checks, so passing them supports, but does not establish, normality.*

6.1 Normal marginals

Proportion check. For variable i and $k = 1, 2$ let $p_{ik} := P(|X_i - \mu_i| < k\sqrt{\sigma_{ii}})$. If $X_i \sim N(\mu_i, \sigma_{ii})$, then $p_{i1} = P(|Z| < 1) = 0.683$ and $p_{i2} = P(|Z| < 2) = 0.954$ for $Z \sim N(0, 1)$. Replace μ_i, σ_{ii} by $\bar{x}_i, s_{ii}$ and count:

$$\hat{p}_{i1} := \frac{\#\{j : |x_{ji} - \bar{x}_i| < \sqrt{s_{ii}}\}}{n}, \qquad \hat{p}_{i2} := \frac{\#\{j : |x_{ji} - \bar{x}_i| < 2\sqrt{s_{ii}}\}}{n}.$$

Flag non-normality if $\hat{p}_{ik}$ is more than three standard errors $\sqrt{p_{ik}(1 - p_{ik})/n}$ away from p_{ik}:

$$|\hat{p}_{i1} - 0.683| > 3\sqrt{\frac{(0.683)(0.317)}{n}} = \frac{1.396}{\sqrt{n}} \quad \text{or} \quad |\hat{p}_{i2} - 0.954| > 3\sqrt{\frac{(0.954)(0.046)}{n}} = \frac{0.628}{\sqrt{n}}.$$

Proportions that are too small suggest tails heavier than normal.

Normal Q–Q plot. If $X \sim N(\mu, \sigma^2)$, its u-quantile is $\mu + \sigma\Phi^{-1}(u)$, linear in $\Phi^{-1}(u)$. The order statistic $x_{(j)}$ estimates the $\frac{j-1/2}{n}$-quantile of X, so with the standard normal quantiles

$$q_{(j)} := \Phi^{-1}\Big(\frac{j - \frac{1}{2}}{n}\Big), \qquad j = 1, \ldots, n,$$

the steps are: (1) order $x_{(1)} \le \cdots \le x_{(n)}$ and compute $q_{(j)}$; (2) plot the pairs $(q_{(j)}, x_{(j)})$; (3) under normality $x_{(j)} \approx \mu + \sigma q_{(j)}$, so check straightness.

Correlation coefficient test. Under normality the Q–Q points lie near a line, so their correlation

$$r_Q := \frac{\sum_{j=1}^n (x_{(j)} - \bar{x})(q_{(j)} - \bar{q})}{\sqrt{\sum_{j=1}^n (x_{(j)} - \bar{x})^2}\sqrt{\sum_{j=1}^n (q_{(j)} - \bar{q})^2}}$$

is near 1. Reject normality at level α if r_Q is below the critical point (Table 4.2 of J&W; excerpt below).

n	$\alpha = .01$	.05	.10
10	.8801	.9198	.9351
20	.9269	.9508	.9604
30	.9479	.9652	.9715
50	.9671	.9768	.9809
100	.9822	.9873	.9895

Example 4.14: Q–Q plot and r_Q (Examples 4.9 and 4.11)

The $n = 10$ ordered observations $-1.00, -0.10, 0.16, 0.41, 0.62, 0.80, 1.26, 1.54, 1.71, 2.30$ have $\bar{x} = 0.770$, $\sqrt{s_{11}} = 0.970$, and $q_{(j)} = \Phi^{-1}((j - \frac{1}{2})/10)$ runs from -1.645 to 1.645. The points

lie close to the line $\bar{x} + \sqrt{s_{11}}\, q = 0.770 + 0.970\, q$ (Figure below), and

$$r_Q = \frac{8.584}{\sqrt{8.472}\,\sqrt{8.795}} = 0.994 > 0.9351,$$

so normality is not rejected at $\alpha = 0.10$.

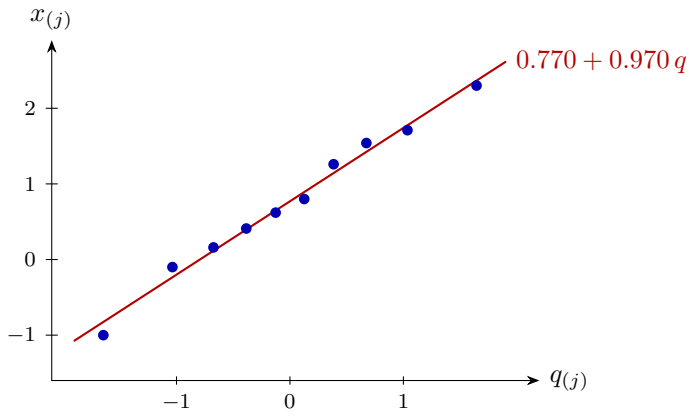


Figure. *Normal Q–Q plot for Example 4.9 ($n = 10$): the ordered data against $q_{(j)} = \Phi^{-1}((j - \frac{1}{2})/n)$. The points are close to the line $\bar{x} + \sqrt{s_{11}}\, q$; $r_Q = 0.994$.*

6.2 Elliptical pairs: the 50% contour

For $p = 2$, $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_2$ by Result 4.7, so the ellipse $\{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \le \chi^2_2(0.5)\}$, $\chi^2_2(0.5) = 1.39$, has probability 0.5. Replacing $\boldsymbol{\mu}, \boldsymbol{\Sigma}$ by $\bar{\mathbf{x}}, \mathbf{S}$, the estimated 50% ellipse $\{\mathbf{x} : (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) \le \chi^2_2(0.5)\}$ should contain about half of the data.

Example 4.15: the 50% contour (Example 4.12)

For the sales–profits data ($n = 10$) the squared Mahalanobis distances are $1.61, 0.30, 0.62, 1.79, 1.30, 4.38, 1.64, 3.53, 1.71, 1.16$. Only 4 of 10 are ≤ 1.39: proportion 0.40 versus 0.50, and $n = 10$ is too small to reject normality.

6.3 Overall fit and outliers: the chi-square plot

$D^2 := (\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$ has u-quantile $q_{c,p}(u)$. Replacing $\boldsymbol{\mu}, \boldsymbol{\Sigma}$ by $\bar{\mathbf{x}}, \mathbf{S}$, the ordered squared distances

$$d^2_j := (\mathbf{x}_j - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}), \qquad j = 1, \ldots, n,$$

estimate the $\frac{j-1/2}{n}$-quantiles of D^2. (1) Order $d^2_{(1)} \le \cdots \le d^2_{(n)}$ and plot the pairs $(q_{c,p}((j - \frac{1}{2})/n),\, d^2_{(j)})$. (2) Compare with the line of slope 1 through the origin.

- Curvature indicates non-normality; a few points far above the line indicate outliers; about half of the d^2_j should be $\le q_{c,p}(0.5)$.
- The d^2_j are neither independent nor exactly χ^2_p, so use the plot as a guide when n and $n - p$ are at least 25–30.

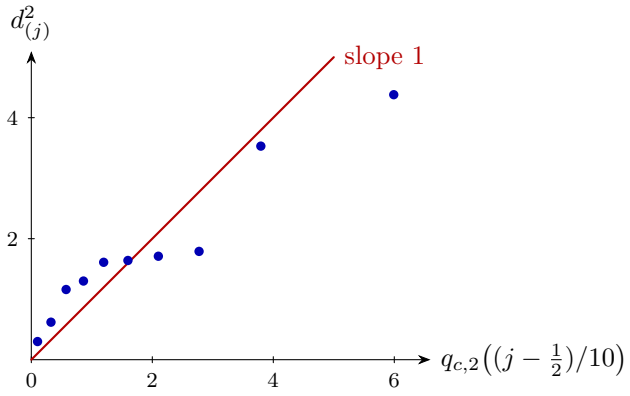


Figure. *Chi-square plot for the sales–profits data (Example 4.13, $p = 2$, $n = 10$), using the distances of Example 4.12. The points follow the 45° line reasonably well, so normality is hard to reject.*

Example 4.16: chi-square plots (Examples 4.13 and 4.14)

(a) Sales–profits ($p = 2$, $n = 10$): the plot above is reasonably straight; it is hard to reject normality.
(b) Board stiffness ($p = 4$, $n = 30$): 17 of the 30 values d_j^2 are $\le q_{c,4}(0.5) = 3.36$ (about half), yet the two largest $d^2_{(j)}$ (boards 9 and 16) break away from the line and the upper end curves, so these boards merit attention.

With small n only gross departures are detected. With large n the tests nearly always reject, but inference on $\boldsymbol{\mu}$ is robust (CLT, Result 4.13).

7 Outliers and Transformations to Near Normality

[Supplement] The slides list J&W Sections 4.7 (detecting outliers and cleaning data) and 4.8 (transformations to near normality) for discussion only; this section summarizes the two tools from J&W.

Outliers. [Supplement] For each variable look at the standardized values $z_{jk} = (x_{jk} - \bar{x}_k)/\sqrt{s_{kk}}$ (dot plots, $|z_{jk}|$ large), at scatter plots of pairs, and at the squared distances d_j^2: an observation with d_j^2 far above the chi-square plot line, or above $\chi^2_p(0.005)$, is a multivariate outlier even if each coordinate looks unremarkable.

Box–Cox transformation. [Supplement] For a positive variable x consider the power family

$$x^{(\lambda)} = \begin{cases} \dfrac{x^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log x, & \lambda = 0, \end{cases}$$

which is continuous in λ. Assuming $x_1^{(\lambda)}, \dots, x_n^{(\lambda)}$ iid normal for some λ, maximizing the likelihood over the mean and variance (profiling, Theorem 4.7 with $p = 1$) and including the Jacobian $\prod_j x_j^{\lambda-1}$ of $x \mapsto x^{(\lambda)}$ gives the profile log-likelihood

$$\ell(\lambda) = -\frac{n}{2} \log\Big[\frac{1}{n}\sum_{j=1}^{n} \big(x_j^{(\lambda)} - \overline{x^{(\lambda)}}\big)^2\Big] + (\lambda - 1)\sum_{j=1}^{n} \log x_j,$$

and $\hat{\lambda} = \arg\max \ell(\lambda)$. By the likelihood ratio, $\{\lambda : \ell(\lambda) \ge \ell(\hat{\lambda}) - \frac{1}{2}\chi^2_1(0.05)\}$ is an approximate 95% interval; in practice a convenient value in it $(0, \frac{1}{2}, 1, \dots)$ is used. For several variables one transforms each marginal separately, which improves but does not guarantee joint normality.

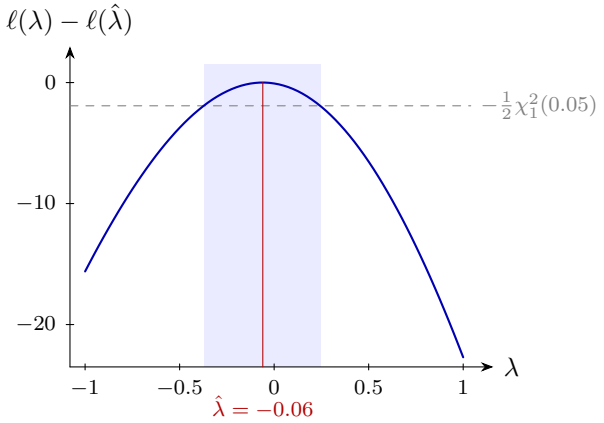


Figure. *[Supplement] Box–Cox profile log-likelihood for a simulated right-skewed sample ($n = 40$, log-normal). The maximum is at $\hat{\lambda} = -0.06$; the shaded band $[-0.37, 0.24]$ is the approximate 95% interval, which contains $\lambda = 0$ (the log transform) but not $\lambda = 1$ (no transformation).*

8 Summary: Distributional Facts Used Later

[Supplement] The facts of Chapters 3 and 4 that the inference chapters rely on:

Fact	Source
$\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$	Theorem 3.8
$(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$, $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$	Theorem 4.3
$(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$	Theorem 3.22, Example 3.23
$\mathbf{a}^\top \boldsymbol{\Sigma}^{-1} \mathbf{a} / \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a} \sim \chi^2_{m-p+1}$, also for independent random $\mathbf{y}$	Theorem 4.5
$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \Rightarrow \chi^2_p$ (no normality)	Theorem 4.11 + Slutsky

9 Selected exercises

Motivation. Two classical consequences of this chapter: the exact law of the generalized variance $|\mathbf{S}|$, obtained from the argument behind Theorem 4.5, and the large-sample law of the sample correlation, obtained from the CLT and the delta method.

Exercise 4.1 (The generalized variance)

Let $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ and $n \geq p$.

(a) Show that $|\mathbf{M}|/|\boldsymbol{\Sigma}| \overset{d}{=} \prod_{i=1}^{p} V_i$ with $V_1, \ldots, V_p$ independent and $V_i \sim \chi^2_{n-i+1}$.

(b) For a sample of size $n > p$ from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, deduce $\mathrm{E}|\mathbf{S}| = |\boldsymbol{\Sigma}| \prod_{i=1}^{p} \dfrac{n-i}{n-1}$, so $|\mathbf{S}|$ underestimates $|\boldsymbol{\Sigma}|$ on average when $p \geq 2$.

Solution [Supplement].

(a) Induction on p. For $p = 1$, $M/\sigma \sim \chi^2_n$ by definition.

Step 1 (set-up). Let $p \geq 2$ and write $\mathbf{M} = \mathbf{X}^\top \mathbf{X}$, where the $n \times p$ matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ has iid rows $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{x}_1$ is its first column. The rows of $\mathbf{X}_2$ are iid $N_{p-1}(\mathbf{0}, \boldsymbol{\Sigma}_{22})$, so $\mathbf{M}_{22} = \mathbf{X}_2^\top \mathbf{X}_2 \sim W_{p-1}(n, \boldsymbol{\Sigma}_{22})$, and $\mathbf{M}_{22} \succ 0$ a.s. since $n \geq p - 1$.

Step 2 (block determinants). By Theorem 2.7, $|\mathbf{M}| = |\mathbf{M}_{22}|\, m_{11\cdot 2}$ and $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{22}|\, \sigma_{11\cdot 2}$, so

$$\frac{|\mathbf{M}|}{|\boldsymbol{\Sigma}|} = \frac{|\mathbf{M}_{22}|}{|\boldsymbol{\Sigma}_{22}|} \cdot \frac{m_{11\cdot 2}}{\sigma_{11\cdot 2}}.$$

Step 3 (the last factor). The proof of Theorem 4.5(i) shows that, conditionally on $\mathbf{X}_2$, $U := m_{11\cdot 2}/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$. This conditional law does not depend on $\mathbf{X}_2$, so $U \sim \chi^2_{n-p+1}$ and U is independent of $\mathbf{X}_2$, hence of $|\mathbf{M}_{22}|/|\mathbf{\Sigma}_{22}|$.

Step 4 (induction). By the induction hypothesis for $\mathbf{M}_{22}$ (dimension $p-1$, $n \geq p-1$), $|\mathbf{M}_{22}|/|\mathbf{\Sigma}_{22}| \overset{d}{=} \prod_{i=1}^{p-1} V_i$ with independent $V_i \sim \chi^2_{n-i+1}$. Put $V_p = U \sim \chi^2_{n-p+1}$; by Step 3 it is independent of $(V_1, \dots, V_{p-1})$.

(b) By Theorem 4.4, $(n-1)\mathbf{S} \sim W_p(n-1, \mathbf{\Sigma})$ with $n-1 \geq p$, so by (a) $(n-1)^p|\mathbf{S}|/|\mathbf{\Sigma}|$ is a product of independent χ^2_{n-i} variables, $i = 1, \dots, p$. Independence gives $\mathrm{E}\{(n-1)^p|\mathbf{S}|/|\mathbf{\Sigma}|\} = \prod_{i=1}^{p}(n-i)$. For $p \geq 2$ the factor $(n-2)/(n-1)$ is < 1 and the others are ≤ 1. ∎

Exercise 4.2 (Asymptotics of the sample correlation)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_2(\boldsymbol{\mu}, \mathbf{\Sigma})$ with $\rho = \sigma_{12}/\sqrt{\sigma_{11}\sigma_{22}}$, $|\rho| < 1$, and $r = s_{12}/\sqrt{s_{11}s_{22}}$.
(a) Show $\sqrt{n}(r-\rho) \Rightarrow N(0, (1-\rho^2)^2)$.
(b) Show that Fisher's $z = \frac{1}{2}\log\frac{1+r}{1-r}$ satisfies $\sqrt{n}(z-\zeta) \Rightarrow N(0,1)$, $\zeta = \frac{1}{2}\log\frac{1+\rho}{1-\rho}$, and give an approximate $1-\alpha$ confidence interval for ρ.

Solution [Supplement].

(a)

Step 1 (reduction). r is unchanged by $X_{ij} \mapsto (X_{ij}-\mu_j)/\sqrt{\sigma_{jj}}$, so assume $\boldsymbol{\mu} = \mathbf{0}$, $\sigma_{11} = \sigma_{22} = 1$, $\sigma_{12} = \rho$.

Step 2 (moments about 0). Let $\mathbf{Y}_i = (X_{i1}^2, X_{i2}^2, X_{i1}X_{i2})^\top$, iid with mean $\boldsymbol{m} = (1, 1, \rho)^\top$, and

$$\mathbf{W}_n = \bar{\mathbf{Y}}_n - (\bar{X}_1^2,\ \bar{X}_2^2,\ \bar{X}_1\bar{X}_2)^\top = \tfrac{n-1}{n}(s_{11}, s_{22}, s_{12})^\top.$$

Since $\sqrt{n}\,\bar{X}_j\bar{X}_k = (\sqrt{n}\bar{X}_j)\bar{X}_k = O_p(1)\,o_p(1) \xrightarrow{p} 0$ (CLT and LLN), Theorem 4.11 and Slutsky give $\sqrt{n}(\mathbf{W}_n - \boldsymbol{m}) \Rightarrow N_3(\mathbf{0}, \mathbf{\Gamma})$ with $\mathbf{\Gamma} = \mathrm{Cov}(\mathbf{Y}_1)$. The factor $\frac{n-1}{n}$ cancels in r, so $r = g(\mathbf{W}_n)$ with $g(a,b,c) = c/\sqrt{ab}$.

Step 3 ($\mathbf{\Gamma}$). Write $X_2 = \rho X_1 + \sqrt{1-\rho^2}\,E$ with $E \sim N(0,1)$ independent of X_1 (Theorem 3.12). Using $\mathrm{E}X_1^4 = 3$ and $\mathrm{E}X_1^3 = 0$,

$$\mathrm{E}X_1^2X_2^2 = 3\rho^2 + (1-\rho^2) = 1 + 2\rho^2, \qquad \mathrm{E}X_1^3X_2 = 3\rho, \qquad \mathrm{E}X_1X_2^3 = 3\rho,$$

and subtracting the products of the means,

$$\mathbf{\Gamma} = \begin{pmatrix} 2 & 2\rho^2 & 2\rho \\ 2\rho^2 & 2 & 2\rho \\ 2\rho & 2\rho & 1+\rho^2 \end{pmatrix}.$$

Step 4 (delta method). g is continuously differentiable near $\boldsymbol{m}$ with $\nabla g(\boldsymbol{m}) = (-\rho/2, -\rho/2, 1)^\top$. The proof of Theorem 4.13 uses only $\sqrt{n}(\mathbf{W}_n - \boldsymbol{m}) \Rightarrow N_3(\mathbf{0}, \mathbf{\Gamma})$ and a Taylor expansion, so $\sqrt{n}(r-\rho) \Rightarrow N(0, \nabla g^\top \mathbf{\Gamma} \nabla g)$ with

$$\begin{aligned} \nabla g^\top \mathbf{\Gamma} \nabla g &= \frac{\rho^2}{4}(\Gamma_{11} + \Gamma_{22} + 2\Gamma_{12}) + \Gamma_{33} - \rho(\Gamma_{13} + \Gamma_{23}) \\ &= \rho^2(1+\rho^2) + 1 + \rho^2 - 4\rho^2 = (1-\rho^2)^2. \end{aligned}$$

(b) $h(x) = \frac{1}{2}\log\frac{1+x}{1-x}$ has $h'(\rho) = 1/(1-\rho^2)$. The one-dimensional delta method applied to (a) gives $\sqrt{n}(z-\zeta) \Rightarrow N\big(0, (1-\rho^2)^2/(1-\rho^2)^2\big) = N(0,1)$: the transformation stabilizes the variance. Hence $P(|z-\zeta| \leq z_{\alpha/2}/\sqrt{n}) \to 1-\alpha$, and since $\tanh = h^{-1}$ is increasing,

$$\big[\tanh(z - z_{\alpha/2}/\sqrt{n}),\ \tanh(z + z_{\alpha/2}/\sqrt{n})\big]$$

is an approximate $1-\alpha$ interval for ρ (in practice $\sqrt{n-3}$ is used, which has the same limit). ∎